

**PROJECT TITLE:**  
**PFSense Firewall Configuration**  
**SUBJECT:**  
**Cyber Security**  
**ASSIGNED BY:**  
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**ASSIGNED TO:**  
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**ROLL NO:**  
**BSIT51F21S053**

## **Introduction:**

What is this Project About?

- Configuring and using PFSense firewall for secure network management.
- Demonstrates how to isolate and control network traffic using virtual machines.
- A step toward understanding real-world network security solutions.

## **PFSense Firewall Overview:**

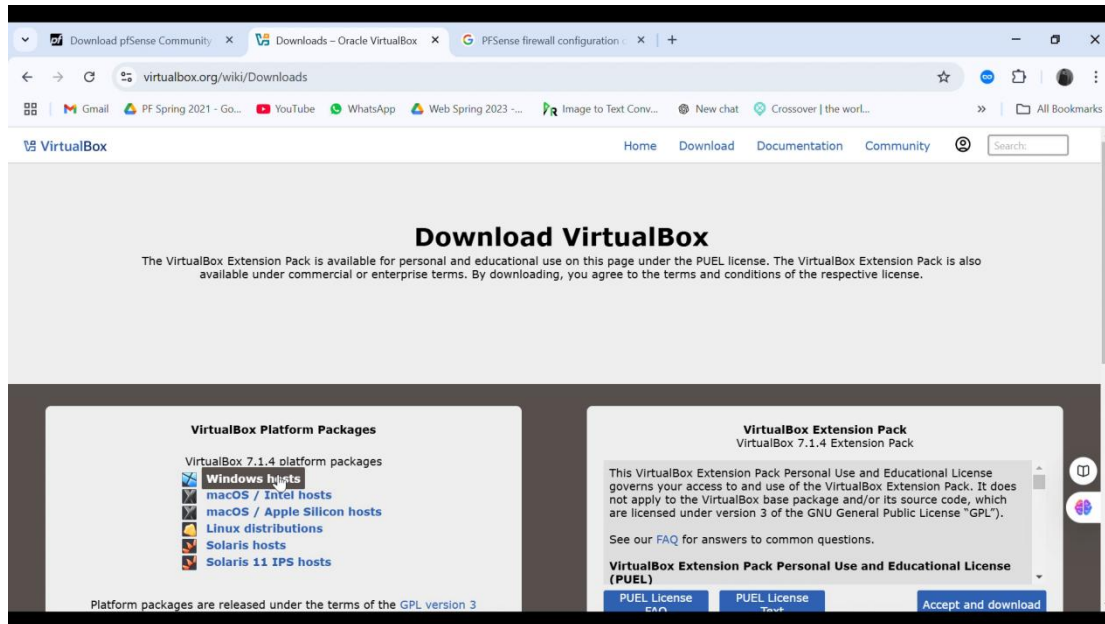
What is PFSense?

- An open-source firewall and router software.
- Features:
  - Traffic Filtering
  - Network Address Translation (NAT)
  - VPN and Proxy Servers
  - Monitoring and Logging

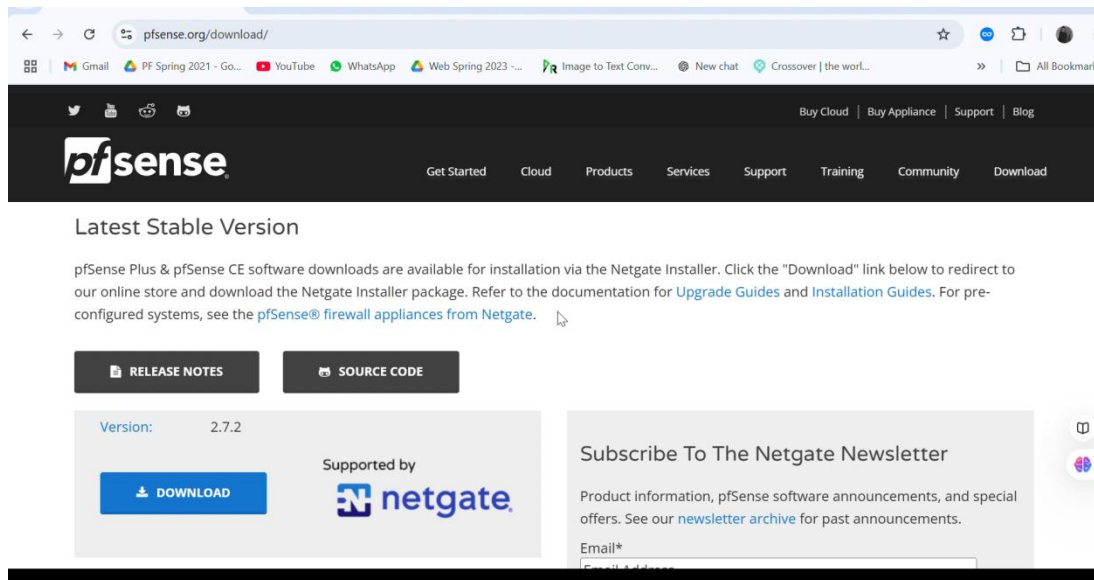
# Virtual Environment Setup:

## Requirements:

- Virtual Box/VMware

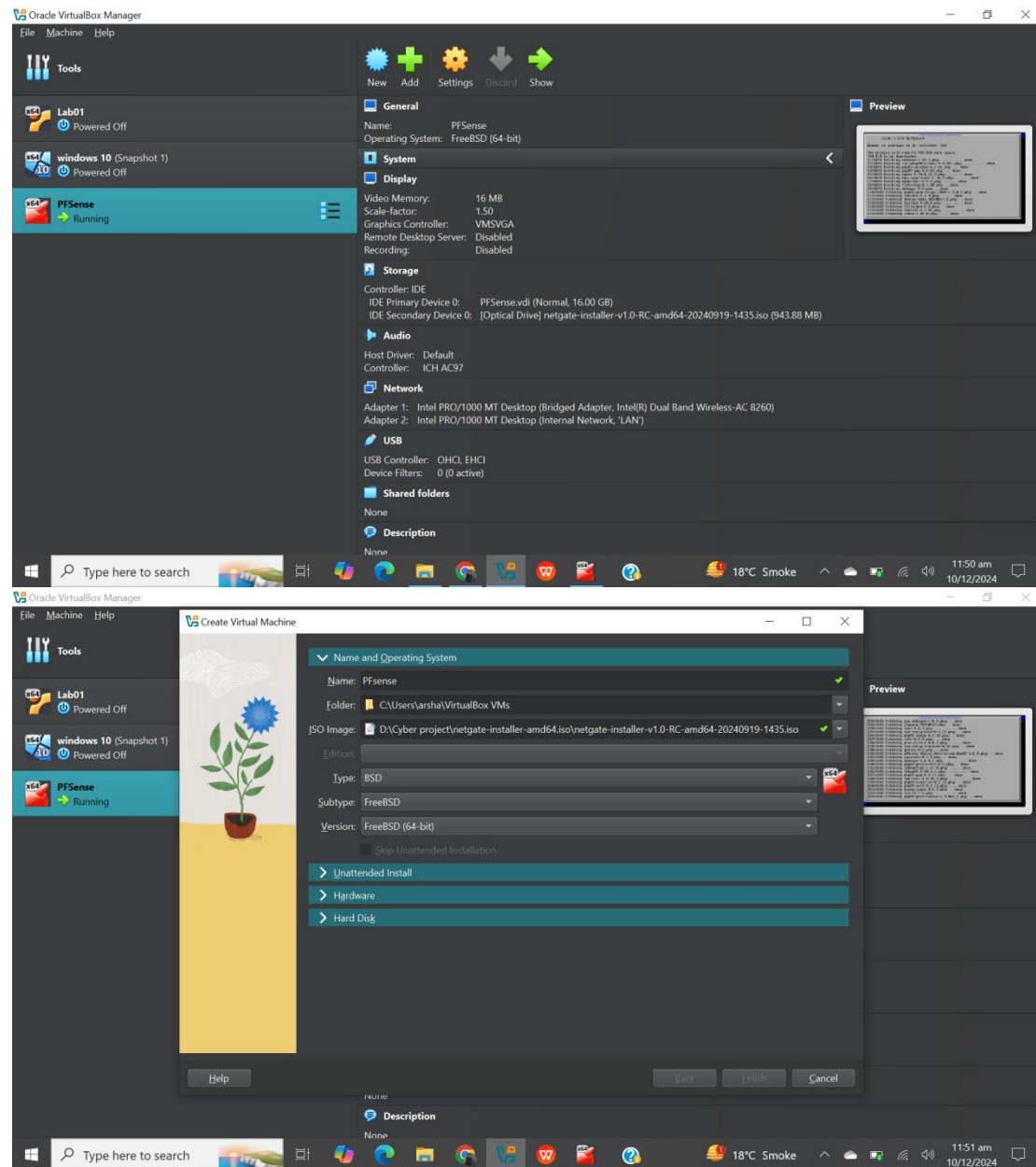


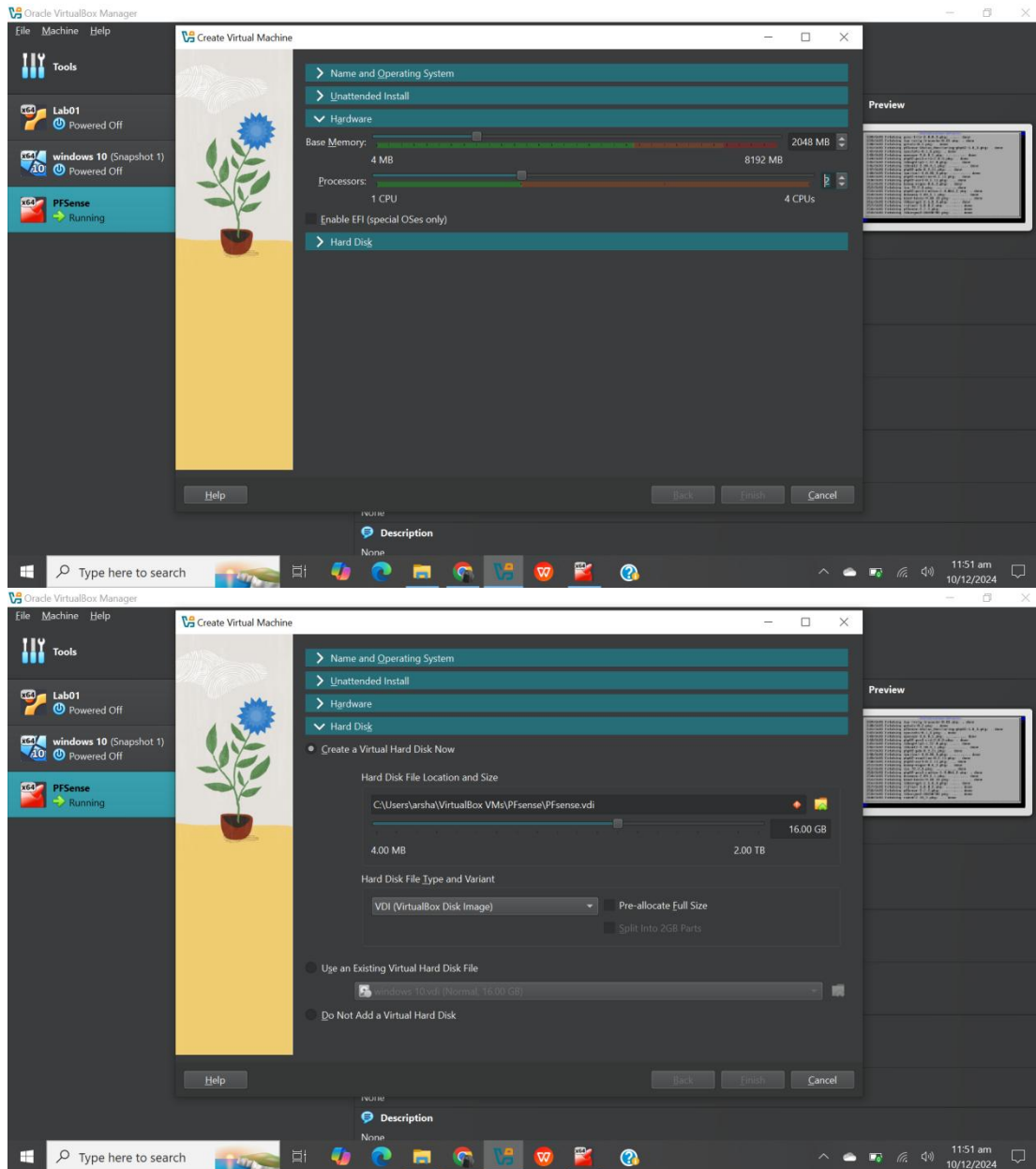
- PFSense ISO Image



# Components in VirtualBox:

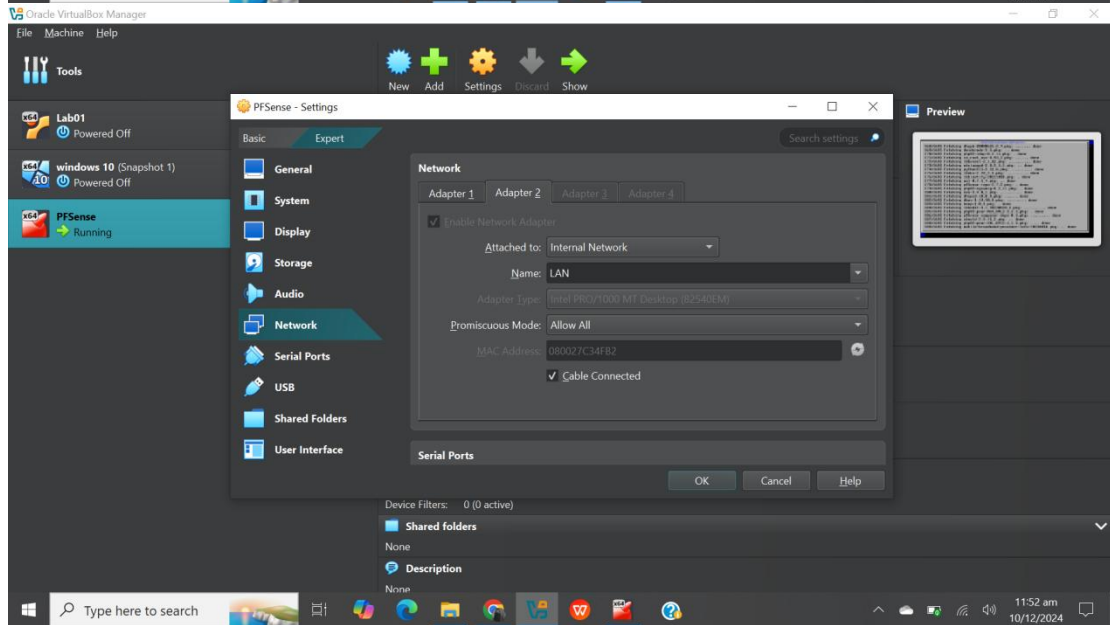
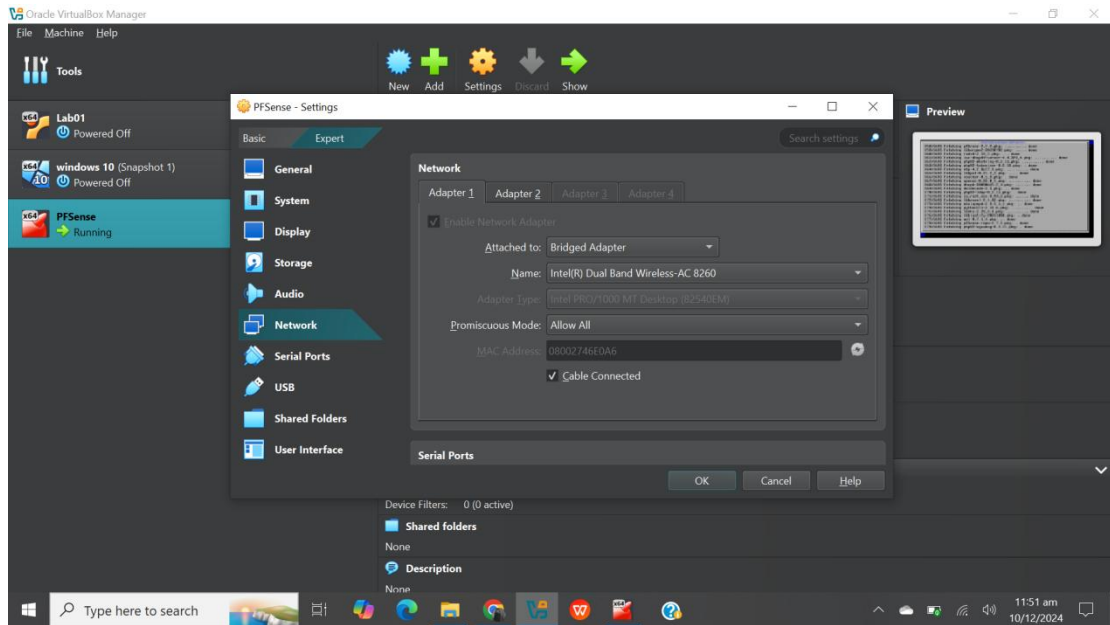
## PFSense VM:



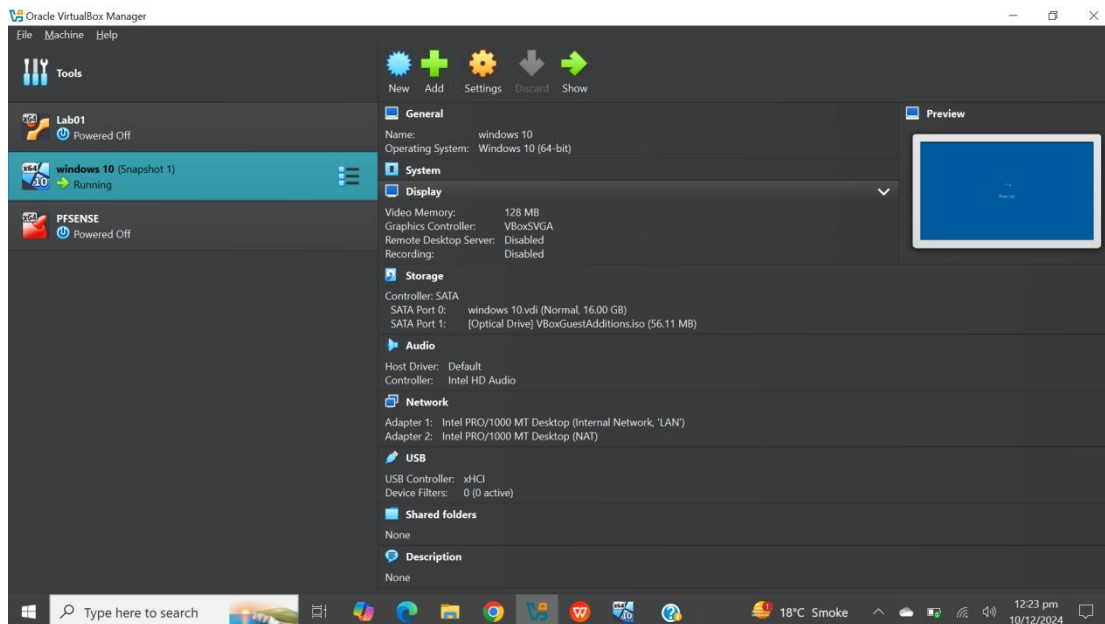


## Network Adapters:

- Interface 1: WAN (Bridged Network).
- Interface 2: LAN (Internal Network).

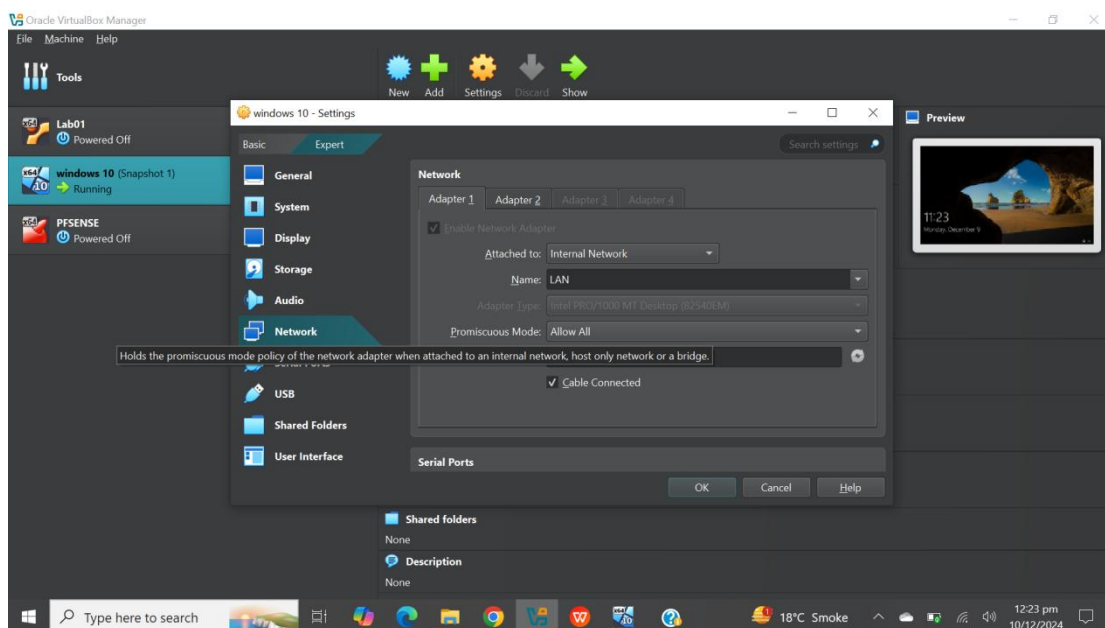


## Windows Host VM



### Network Adapters:

- Connected to LAN Internal Network.

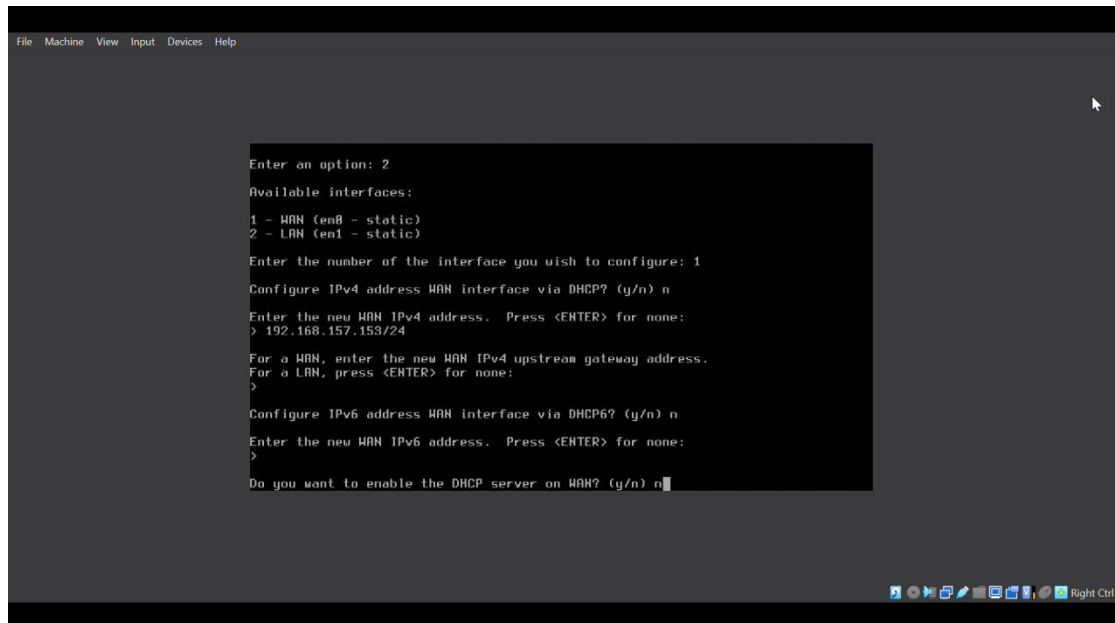


- After all the settings of network adapters click to start the machine and by continue with default settings setup pfSense.

# PFSense VM's Network Configuration:

## WAN Interface (PFSense):

- IP Address: 192.168.157.153/24.



The screenshot shows a terminal window with a menu bar at the top containing 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help'. The terminal text is as follows:

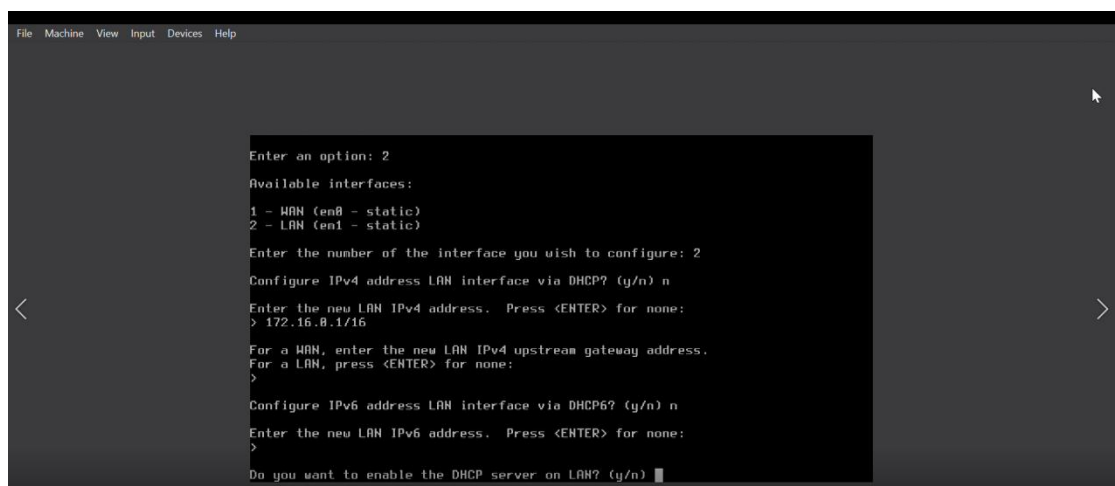
```
Enter an option: 2
Available interfaces:
1 - WAN (en0 - static)
2 - LAN (en1 - static)

Enter the number of the interface you wish to configure: 1
Configure IPv4 address WAN interface via DHCP? (y/n) n
Enter the new WAN IPv4 address. Press <ENTER> for none:
> 192.168.157.153/24
For a WAN, enter the new WAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>
Configure IPv6 address WAN interface via DHCP6? (y/n) n
Enter the new WAN IPv6 address. Press <ENTER> for none:
>
Do you want to enable the DHCP server on WAN? (y/n) n
```

At the bottom right of the terminal window, there is a taskbar with various icons and the text 'Right Ctrl'.

## LAN Interface (PFSense):

- IP Address: 172.16.0.1/16.



The screenshot shows a terminal window with a menu bar at the top containing 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help'. The terminal text is as follows:

```
Enter an option: 2
Available interfaces:
1 - WAN (en0 - static)
2 - LAN (en1 - static)

Enter the number of the interface you wish to configure: 2
Configure IPv4 address LAN interface via DHCP? (y/n) n
Enter the new LAN IPv4 address. Press <ENTER> for none:
> 172.16.0.1/16
For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>
Configure IPv6 address LAN interface via DHCP6? (y/n) n
Enter the new LAN IPv6 address. Press <ENTER> for none:
>
Do you want to enable the DHCP server on LAN? (y/n) n
```

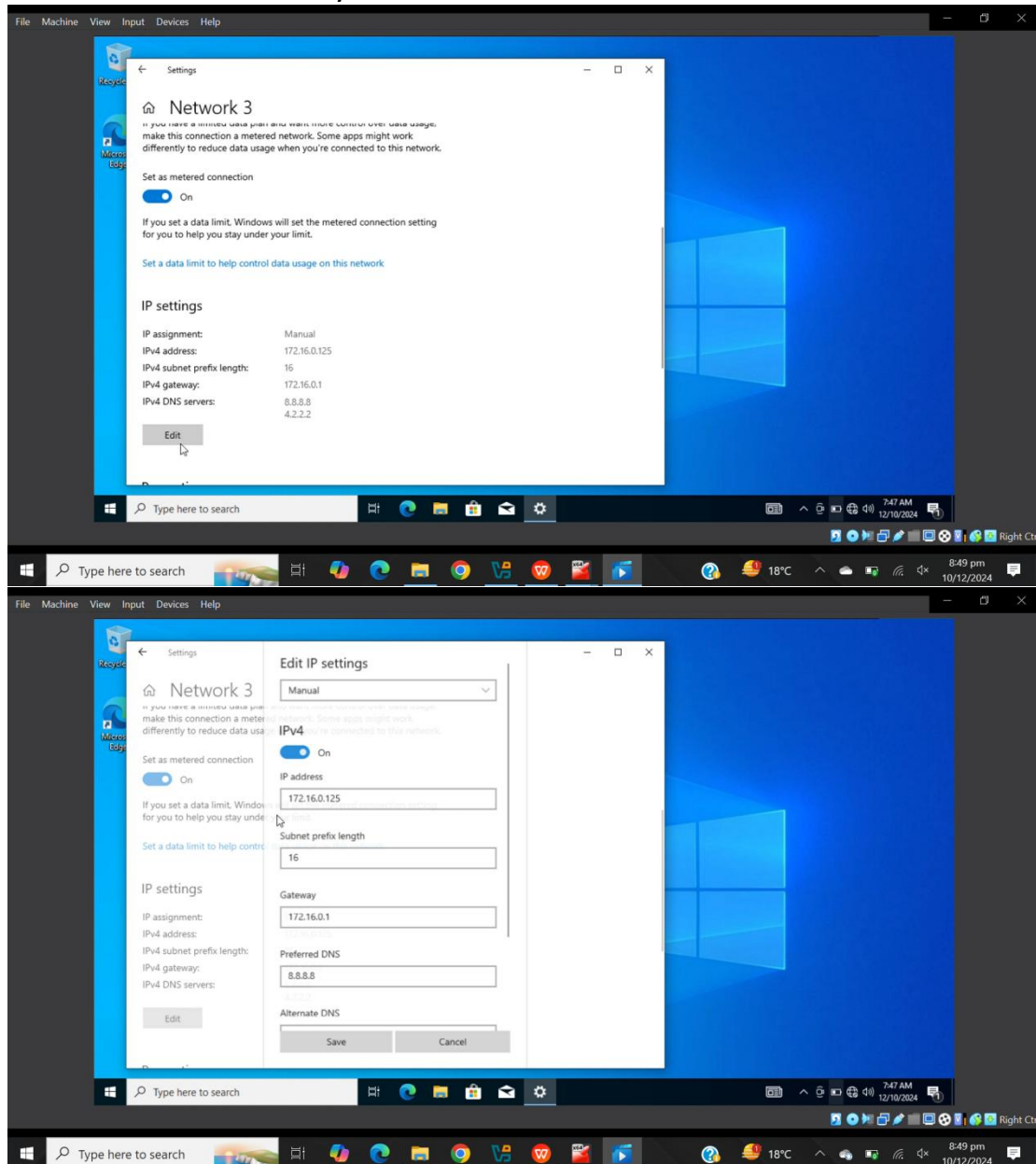
On the left and right sides of the terminal window, there are navigation arrows: a left-pointing arrow on the left and a right-pointing arrow on the right.

# Windows Host VM's Network Configuration:

Steps:

➤ Configure:

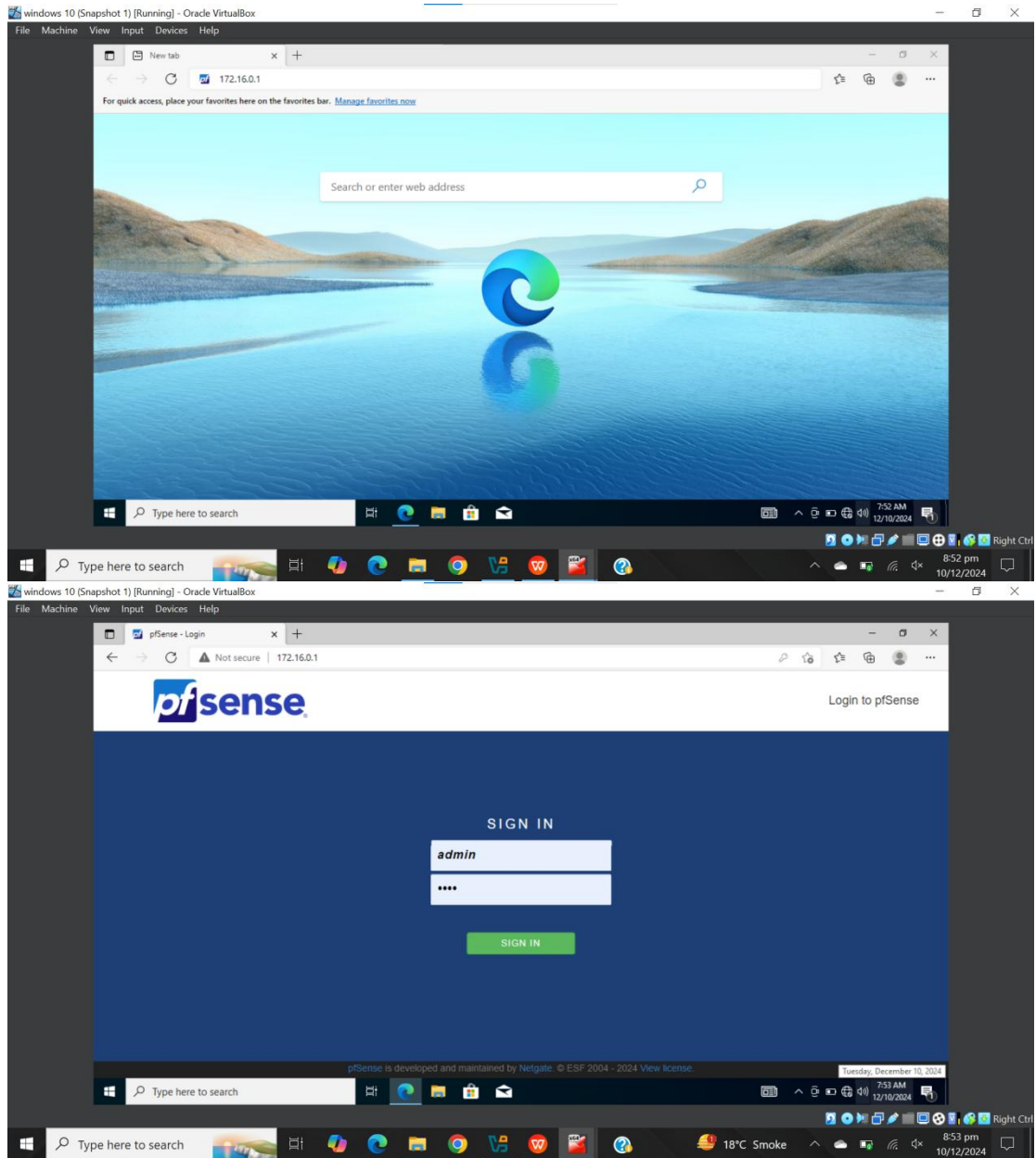
- IP Address: 172.16.0.125
- Subnet Mask: 255.255.0.0
- Default Gateway: 172.16.0.1

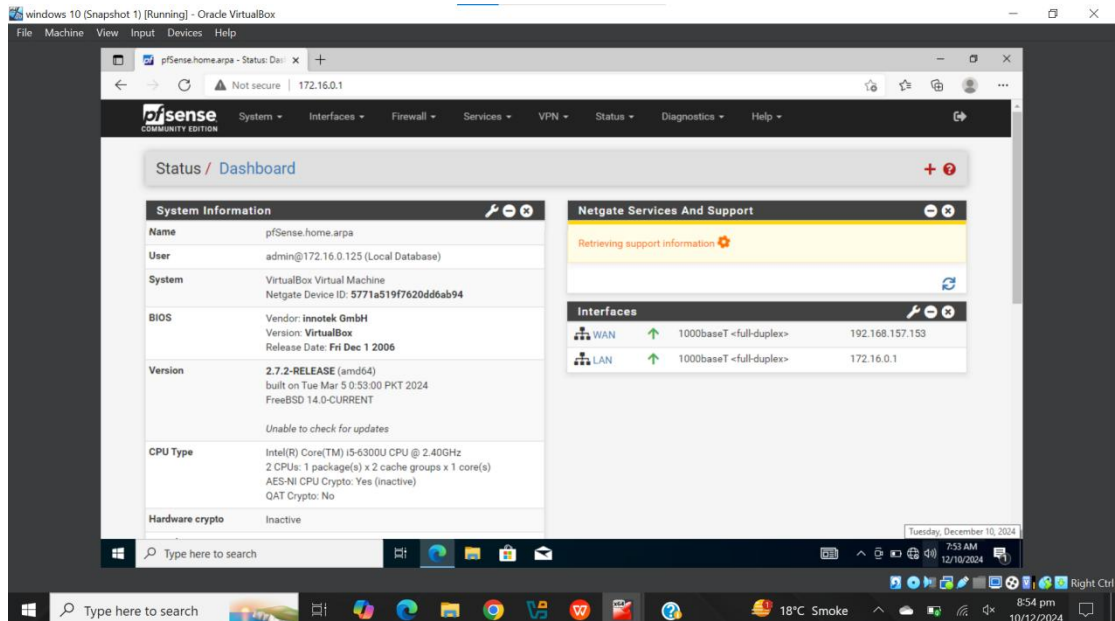


## Accessing PFSense GUI:

Steps to Access:

- Open a browser on Windows Host VM.
- Enter the LAN IP: 172.16.0.1.
- Login to PFSense with default credentials (admin/pfsense).





## PFsense Firewall Rules:

### Rule to block facebook.com

Firewall>Rules>Add Rule to LAN Interface

Action: Block

Protocol: TCP/UDP

Source: LAN Subnets

Destination: Any

Destination Port Range: 80 for HTTP - 443 for HTTPS

Description: Facebook Blocked

Save and click to apply changes

### Rule by DNS Resolver:

Click add under Host Override option.

Hostname: Facebook

Domain Name: facebook.com

IP Address: 0.0.0.0

Description: Facebook Blocked

Click save and apply changes.

Test the Rules by using pfsense shell or DNS Lookup.

## **Purpose and Benefits of PFSense:**

Why Use PFSense?

- Security: Filters and controls traffic.
- Access Management: Restricts unauthorized users.
- Network Segmentation: Creates isolated networks.
- Monitoring: Logs and monitors traffic for analysis.

Real-Life Use Cases:

- Office network security.
- Educational institution internet management.
- Secure home networks.